

Working with Matrices

Let **A**, **B**, and **C** be the matrices given below.

$$A = \begin{bmatrix} 1 & -2 & 3 & 7 \\ 2 & 1 & 1 & 4 \\ -3 & 2 & -2 & 10 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 & -3 & -2 \\ 10 & -1 & 2 & -3 \\ 5 & 1 & -1 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 4 & 0 & -2 & 3 \\ 3 & 6 & 9 & 7 \\ 2 & 2 & 5 & 1 \\ 9 & 4 & 6 & -2 \end{bmatrix}$$

1. Enter each of the matrices A , B , and C as Numpy matrix.
2. What are the dimensions of each of the matrices A , B , and C ? Display these using Python code.
3. Determine if each of the following matrix operations is defined. If it is defined, calculate the result using Python. If it is not defined, explain why it is not defined.
 - (a) $A + B$
 - (b) AB
 - (c) AC
 - (d) C^T
4. Illustrate the following property of matrix transpositions using these matrices: $(A + B)^T = A^T + B^T$.
5. Illustrate the following property of matrix transposition using these matrices $(AC)^T = C^T A^T$.
6. Find C^{-1} using Python code.

Working with Images

For this exercise, you will be working with the image file **week7_image.jpg**.

1. Use the following code to display the image.

```
#import the OpenCV library to read images
import cv2
#import matplotlib to display the image
import matplotlib.pyplot as plt

#import the image; by default, OpenCV imports images in the BGR format
image_bgr = cv2.imread('week7_image.jpg', cv2.IMREAD_COLOR) #you may need to modify your file path
#convert the image to the RGB format
image_rgb = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2RGB)

#display the image
plt.imshow(image_rgb)
plt.axis("off")
plt.show()
```

2. Find the resolution of this image. **Hint:** OpenCV imports the image as a Numpy array, where each pixel corresponds to an entry in the array.
3. In checking the array dimensions, you should see that three numbers are displayed. What is this third number, and why is it there?